CLASS 12 | ECONOMICS | CUET

Macroeconomics Formula Sheet

17 pages | every formula, table and diagram | clay style

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Source notes: abhipedia CUET topper notes (Milap Bansal), redesigned in the wondermayank style.

NATIONAL INCOME | BASICS

National Income: General Conversions

01

Quick conversions between gross, net, market price, factor cost and the income concepts.

Price and depreciation conversions

Gross = Net + Depreciation

Market price (MP) = Factor cost (FC) + Net indirect tax (NIT)

NIT = Indirect taxes - Subsidies

Basic price (BP) = FC + Net production taxes

National = Domestic + Net factor income from abroad (NFIA)

NFIA = Factor income received from abroad - Factor income paid to abroad

Personal and national income concepts

PI Personal Income (PI) = NI - UP - NIP - CT + TPH

UP: undistributed profits | NIP: net interest payments | CT: corporate tax | TPH: transfer payments received by households / firms from govt.

PDI Personal Disposable Income (PDI) = PI - PTP - NTP

PTP: personal tax payments | NTP: personal non-tax payments

NDI National Disposable Income (NDI) = NNP_{MP} + Net current transfers from abroad

Pvt Y Private income = Factor income from NDP accruing to private sector + NFIA + current transfers from govt. + current transfers from the rest of the world

QUICK TIP

Gross to net: subtract depreciation. Market price to factor cost: subtract NIT. Domestic to national: add NFIA.

NATIONAL INCOME | METHODS

GDP Deflator and Value Added Method

02

One formula in three forms, then the first of the three methods of calculating national income.

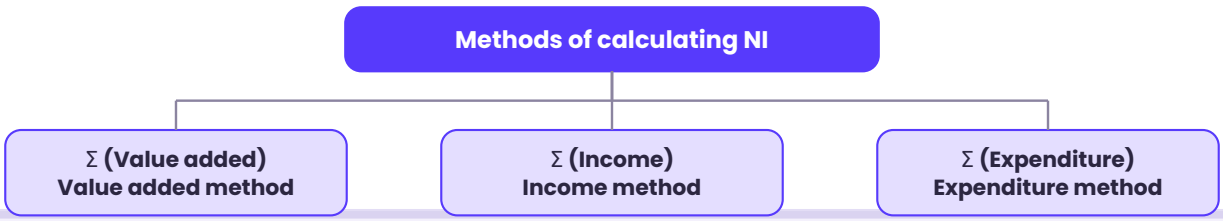
GDP deflator

GDP deflator = (Nominal GDP ÷ Real GDP) × 100

Nominal GDP = (GDP deflator × Real GDP) ÷ 100

Real GDP = (Nominal GDP ÷ GDP deflator) × 100

Methods of calculating national income (NI)



Value added method

Step 1: $GDP_{MP} = VO - IC$ or $GDP_{MP} = \Sigma GVA_i$

GDP_{MP} = gross domestic product at market price | VO = value of output | IC = intermediate consumption

VO = Sales + ΔStock

Sales = domestic sales + exports. If 'sales' is given, do not add exports again!

IC = Domestic purchases + Imports of intermediate goods

Use this only if domestic purchases are given.

Step 2: $NNP_{FC} = GDP_{MP} - NIT + NFIA - Depreciation$

NATIONAL INCOME | METHODS

Income Method

03

Add up all factor incomes: COE, rent, interest, profits and mixed income.

Components of domestic income (NDP at factor cost)

COE	+	RENT	+	INTEREST	+	PROFITS	+	MI
Compensation of employees		Rent and royalty		Interest		Profits		Mixed income
• Wages and salaries • Payments in kind • Employer's contribution to social security schemes (not employee's) • Retirement pensions (not old-age pension)		• Rent including actual and imputed rent		• Net interest • Counted only if the loan is for productive purposes		• Dividends (distributed profits) • Undistributed profits • Corporate profit tax • All three are part of profits		• Self-employed income

Operating surplus (OS) = r + i + p (rent + interest + profits)

Formulas

Step 1: NDP_{FC} (domestic income) = COE + OS + MI

COE = compensation of employees | OS = operating surplus (r + i + p) | MI = mixed income

Step 2: NNP_{FC} (national income) = NDP_{FC} + NFIA

CROSS-CHECK

All three methods must give the same answer. Link between them: GDP_{MP} = NDP_{FC} + Depreciation + NIT.

NATIONAL INCOME | METHODS

Expenditure Method

04

Add up all spending on final goods: PFCE + GFCE + GDCF + NX.

Components of GDP at market price

PFCE	+	GFCE	+	GDCF	+	NX
Private final consumption expenditure		Government final consumption expenditure		Gross domestic capital formation		Net exports
Exception: when households buy a new house, it is counted as investment (GDCF), not PFCE		Does not include transfer payments		GDCF = GDFCF + ΔStock GDFCF = gross domestic fixed capital formation		NX = Exports - Imports = X - M PFCE, GFCE and GDCF also include spending on imports, so imports are subtracted through NX

Formulas

Step 1: GDP_{MP} = PFCE + GFCE + GDCF + NX

NX = net exports = X - M

Step 2: NNP_{FC} = GDP_{MP} - Depreciation - NIT + NFIA

REMEMBER

Government final consumption does not include transfer payments (pensions, scholarships). A household buying a new house is counted as investment, not consumption.

MONEY AND BANKING

Money Supply, Multiplier and Money Demand

05

M0 to M4, the money multiplier, transaction demand and the bond price link.

Measures of money supply

M0 M0 (monetary base / high-powered money) = Currency held by public + Bank reserves with RBI

M1 M1 (narrow money) = Currency (notes and coins) held by public + Net demand deposits with commercial banks

M2 M2 = M1 + Savings deposits with post office savings banks

M3 M3 = M1 + Net time deposits of commercial banks

M4 M4 (broad money) = M3 + Total deposits with post office savings organisations (excluding NSCs)

Money multiplier

m = 1 ÷ CRR (CRR = cash reserve ratio)

m = Total created deposit ÷ Initial deposit and CRR = Initial deposit ÷ Total created deposit

Demand for money (CUET) and bond price

$M^d_T = k \cdot P \cdot Y$

Transaction demand for money | P = price level | Y = real GDP

Price of a bond $\propto$ 1 ÷ Market rate of interest

Inverse relation: interest rate up, bond price down.

Demonetisation: case study given in NCERT (Box 3.2, p. 49)

Important for CUET-UG.

MONEY AND BANKING

Monetary Policy Tools (1 of 2)

06

Rates and ratios that the RBI uses to control credit and liquidity.

Rate and ratio tools

Tool	Definition	Purpose / usage	Impact on economy
Bank Rate	The interest rate at which the RBI lends to commercial banks without collateral.	Used to regulate **long-term credit** in the economy.	↑ Increase → Reduces borrowing, curbs inflation. ↓ Decrease → Encourages borrowing, boosts growth.
Repo Rate	The interest rate at which the RBI lends **short-term** funds to commercial banks against government securities.	Used to **inject liquidity** into the banking system when there is a **shortage** of funds.	↑ Increase → Costlier borrowing for banks, reduces money supply. ↓ Decrease → Cheaper borrowing, increases money supply.
Reverse Repo Rate	The rate at which the RBI **borrows** money from commercial banks.	Used to **absorb excess liquidity** from the system.	↑ Increase → Banks park more money with RBI, reduces money in circulation. ↓ Decrease → Banks lend more, increasing liquidity.
Cash Reserve Ratio (CRR)	The percentage of net demand and time liabilities (NDTL) that banks must keep with the RBI in cash.	Used to control **liquidity** and ensure **bank stability**.	↑ Increase → Reduces funds for lending, lowers money supply. ↓ Decrease → Frees up funds for loans, increases money supply.
Statutory Liquidity Ratio (SLR)	The percentage of NDTL that banks must maintain in the form of liquid assets (cash, gold, approved securities).	Used to control **credit expansion** and ensure bank **solvency**.	↑ Increase → Limits bank lending, tightens money supply. ↓ Decrease → Increases bank lending capacity, boosts liquidity.

QUICK MEMORY

Repo = RBI **lends** to banks (injects money). Reverse repo = RBI **borrows** from banks (absorbs money). When rates and ratios go up, money gets tight and inflation falls; when they go down, money gets loose and growth rises.

MONEY AND BANKING

Monetary Policy Tools (2 of 2)

07

Market operations, standing facilities and the committee that sets the policy rate.

Market, facility and other tools

Tool	Definition	Purpose / usage	Impact on economy
Open Market Operations (OMO)	Buying and selling of **government securities** by the RBI in the **open market**.	Used to manage **liquidity** in the economy.	**Buy securities** → Injects liquidity, increases money supply. **Sell securities** → Absorbs liquidity, reduces money supply.
Liquidity Adjustment Facility (LAF)	A tool through which banks can **borrow from** or **deposit funds** with the RBI (includes Repo and Reverse Repo).	Used to manage **short-term liquidity** fluctuations.	**Repo** → Provides liquidity to banks. **Reverse Repo** → Absorbs excess liquidity.
Marginal Standing Facility (MSF)	Allows banks to borrow **overnight** from the RBI at a rate **higher than the Repo Rate**.	Used as a **last resort** for banks facing **liquidity crises**.	↑ Increase → Costlier emergency borrowing, controls excess money. ↓ Decrease → Cheaper access, increases liquidity.
Monetary Policy Committee (MPC)	A **6-member** committee responsible for setting the **policy interest rate** (Repo Rate).	Ensures **price stability** and **economic growth**.	Targets **inflation** while supporting **economic growth**.
Moral Suasion	**Non-statutory** persuasion by the RBI through meetings and communications.	Used to **influence** banks' credit policies without direct mandates.	Can **encourage / discourage** lending and tighten / loosen liquidity indirectly.
Market Stabilization Scheme (MSS)	Issuance of **bonds and securities** to absorb **excess liquidity**.	Used to neutralize excess liquidity from **foreign capital inflows**.	Reduces **inflationary pressures** by absorbing surplus liquidity.

QUICK MEMORY

OMO: **buy = inject** money, **sell = absorb** money. MSF rate is always above the repo rate. MPC has **6 members** and sets the repo rate.

INCOME AND EMPLOYMENT

Propensities and Investment Function

08

MPC, APC, MPS and APS, plus the investment function.

Propensities to consume and save

MPC $MPC = \Delta C \div \Delta Y$

Change in consumption ÷ change in income. Always between 0 and 1.

APC $APC = C \div Y$

Can never be negative or 0. Equal to 1 when C = Y (more than 1 when C is greater than Y).

MPS $MPS = \Delta S \div \Delta Y$

Change in savings ÷ change in income. Between 0 and 1 (can also equal 0 or 1).

APS $APS = S \div Y$

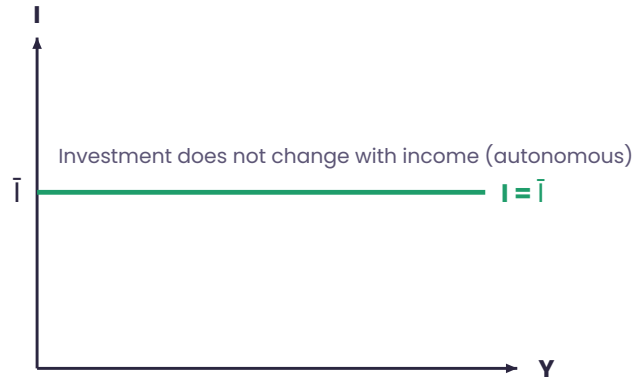
Can be negative (dissaving), zero or positive. Also remember: MPC + MPS = 1 and APC + APS = 1.

Investment function

$I = \bar{I}$

Autonomous investment: it stays the same at every level of income, so the line is horizontal.

Investment function graph



INCOME AND EMPLOYMENT

Consumption and Savings Function

09

The two straight-line functions and the break-even point where C = Y.

Consumption function

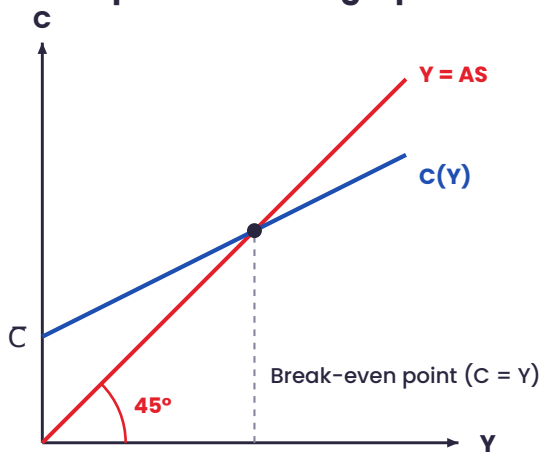
$C(Y) = \bar{C} + c \cdot Y$

$\bar{C}$ = autonomous consumption

$c \cdot Y$ = induced consumption

c = MPC and 1 - c = MPS

Consumption function graph



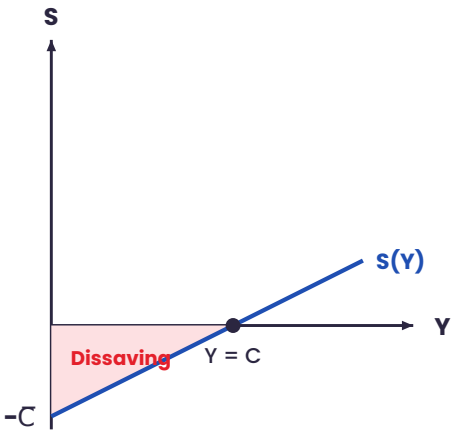
Savings function

$S(Y) = -\bar{C} + Y(1 - c)$

MPS = (1 - c)

Savings are negative (dissaving) below the break-even point, where Y = C

Savings function graph



LINK BETWEEN THEM

The savings line cuts the x-axis at the same income where the consumption line cuts the 45° line. At that break-even point C = Y and savings are zero.

INCOME AND EMPLOYMENT

AD, AS and Equilibrium

10

Equilibrium income Y* is where AD meets the 45° AS line.

Aggregate demand and aggregate supply

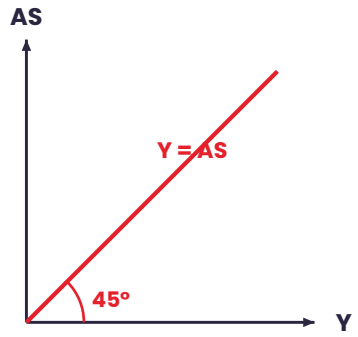
$AD = \bar{C} + \bar{I} + c \cdot Y$

c = MPC

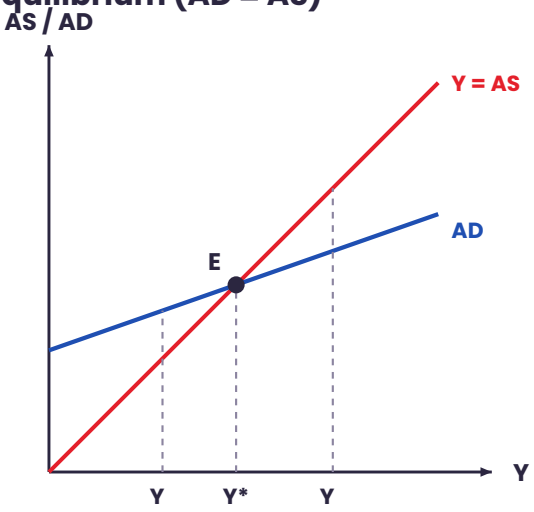
AS: $Y = AS$

The 45° line: whatever is produced is the income earned.

Aggregate supply



Equilibrium (AD = AS)



Adjustment towards equilibrium

If Y is less than Y* ⇒ AS is less than AD ⇒ Y rises towards Y*

Demand is greater than supply, so output rises.

If Y is greater than Y* ⇒ AS is greater than AD ⇒ Y falls towards Y*

Supply is greater than demand, so output falls.

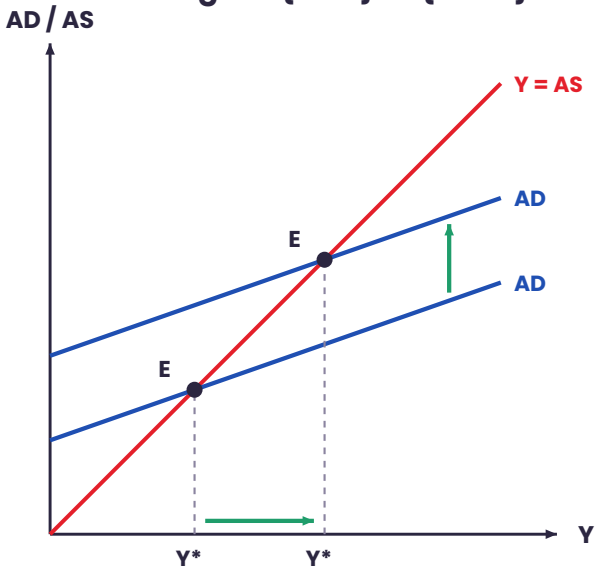
INCOME AND EMPLOYMENT

Shifts in AD and Multipliers

11

What happens to Y* when AD shifts, and how big the change is.

Positive change in {Ibar} or {Cbar}: AD shifts up, Y* rises



RULE

A positive $\Delta\bar{I}$ or $\Delta\bar{C}$ shifts AD up and raises Y*. The reverse happens for a negative $\Delta\bar{I}$ or $\Delta\bar{C}$: AD shifts down and Y* falls.

Investment multiplier

$$k = \Delta Y \div \Delta\bar{I}$$

Change in income ÷ change in investment

$$k = 1 \div (1 - MPC) = 1 \div MPS$$

Government spending multiplier

$$\Delta Y = [1 \div (1 - MPC)] \cdot \Delta G$$

$$k_G = 1 \div (1 - MPC) = 1 \div MPS = \Delta Y \div \Delta G$$

GOVERNMENT BUDGET AND THE ECONOMY

Government Budget: Components

12

Revenue budget, capital budget, budget types and the three measures of deficit.

Budget at a glance

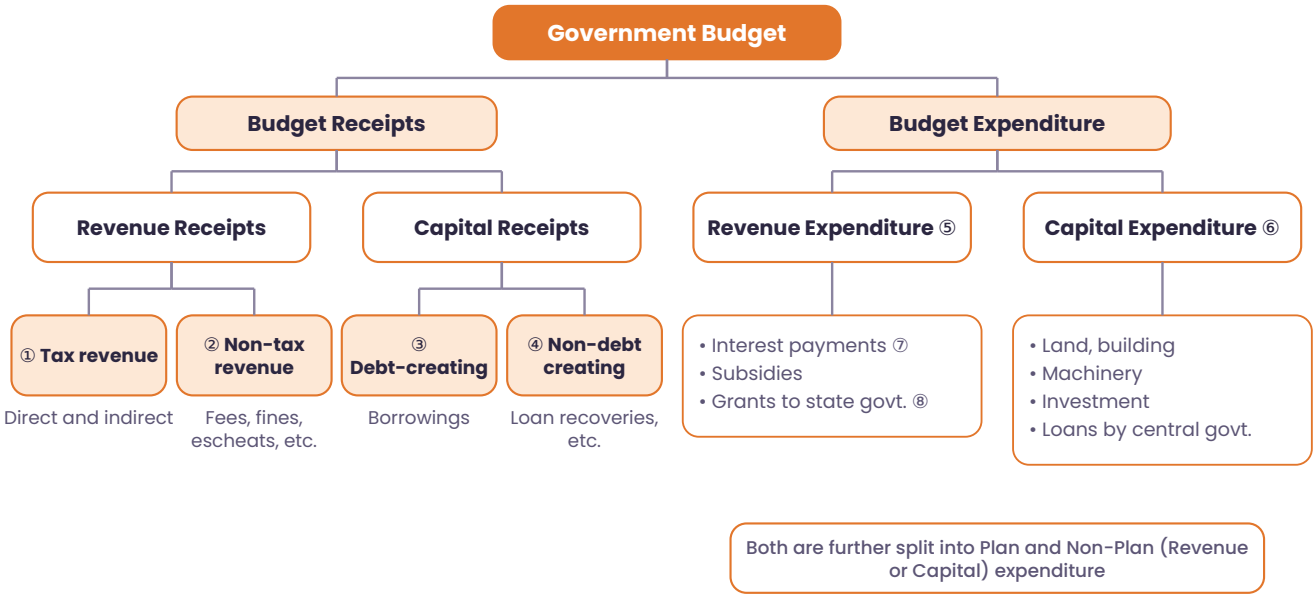
Category	Sub-category	Description
Revenue Budget	**Revenue receipts**	Income received by the government without increasing liabilities.
	- Tax revenue	Includes direct taxes (income tax, corporate tax) and indirect taxes (GST, excise duty).
	- Non-tax revenue	Earnings from interest, dividends, fees, fines, and grants from foreign bodies.
	Revenue expenditure	Expenses for daily functioning, which do not create physical or financial assets.
	- Plan revenue expenditure	Spending on government plans (Five-Year Plans, assistance to states). The plan / non-plan split was removed from the Union Budget in 2017, but many books still use it.
	- Non-plan revenue expenditure	Salaries, pensions, interest payments and subsidies.
Capital Budget	**Capital receipts**	Funds raised through borrowings or asset sales; they create liabilities or reduce assets.
	- Debt-creating capital receipts	Loans that need to be repaid (market borrowings, external loans).
	- Non-debt capital receipts	Proceeds from PSU disinvestment or loan recoveries.
	Capital expenditure	Spending on creating physical / financial assets (infrastructure, machinery, land).
	- Plan capital expenditure	Investment in planned development projects.
	- Non-plan capital expenditure	Spending on assets outside planned schemes (defence, social services).
Budget Types	**Balanced budget**	Government's revenue equals its expenditure.
	Surplus budget	Revenue exceeds expenditure; used to control inflation.
	Deficit budget	Expenditure exceeds revenue; requires borrowing.
Measures of Deficit	**Revenue deficit**	Revenue expenditure exceeds revenue receipts.
	Fiscal deficit	Total expenditure exceeds total receipts (excluding borrowings).
	Primary deficit	Fiscal deficit excluding interest payments on past debt.

Budget Diagram and Deficits

13

Numbers ① to ⑧ on the tree are used in the deficit formulas below.

Government budget tree



Deficit formulas

RD Revenue deficit = Revenue expenditure - Revenue receipts

R.D. = ⑤ - (① + ②)

FD Fiscal deficit = Total expenditure - Total receipts (excluding borrowing)

F.D. = (⑤ + ⑥) - (① + ② + ④) = ③ (the borrowings)

PD Primary deficit = Fiscal deficit - Interest payments

P.D. = F.D. - ⑦

BALANCE OF PAYMENTS

Balance of Payments: Components

14

Current account, capital account, errors and omissions, and official reserves.

BoP at a glance

Category	Sub-category	Description	Examples
1. Current Account	**A. Merchandise trade** (balance of trade)	Records **exports and imports of goods**.	Export of machinery, import of crude oil.
	B. Invisibles	Includes **services, income and transfers**.	Software exports, remittances, tourism.
	- Non-factor services	Services exchanged across borders.	IT services, transport, banking.
	- Income	Earnings from investments abroad.	Dividends, profits, interest.
	- Transfers	**Unilateral transfers** without compensation.	Foreign remittances, gifts, grants.
2. Capital Account	**A. Foreign investment**	Investment flows into and out of the country.	FDI, FII.
	B. External assistance (net)	Loans and grants from foreign governments.	World Bank loans, IMF assistance.
	C. External commercial borrowings (net)	Corporate borrowing from foreign lenders.	Indian company issuing bonds abroad.
	D. Short-term debt	Debt with a maturity of **less than one year**.	Short-term trade credit.
	E. Banking capital	Foreign currency deposits in Indian banks.	NRI deposits.
3. Errors and Omissions	**Statistical discrepancy**	Adjustment for unrecorded or misclassified transactions.	Data errors in international transactions.
4. Official Reserve Transactions	**Foreign exchange reserve**	RBI intervention to balance BoP.	RBI buying / selling foreign currency.

REMEMBER

Current account = goods + services + income + transfers. Capital account = investment + borrowings + assistance. Official reserves bring BoP back to zero.

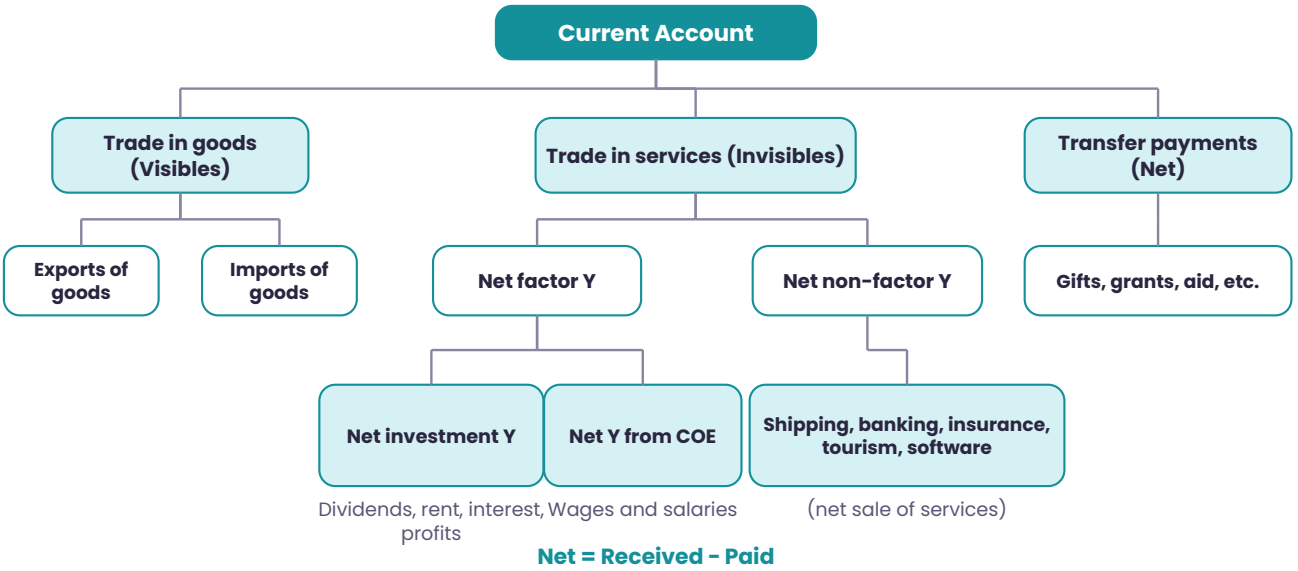
BALANCE OF PAYMENTS

Current Account

Trade in goods, trade in services and transfer payments.

15

Current account tree



Current account balance

Current A/C surplus: Receipts on current A/C are greater than payments on current A/C

Current A/C deficit: Receipts on current A/C are less than payments on current A/C

NOTE

Trade in goods is called 'visibles'; trade in services is called 'invisibles'. Net factor income = income from abroad (dividends, interest, wages) minus income paid abroad.

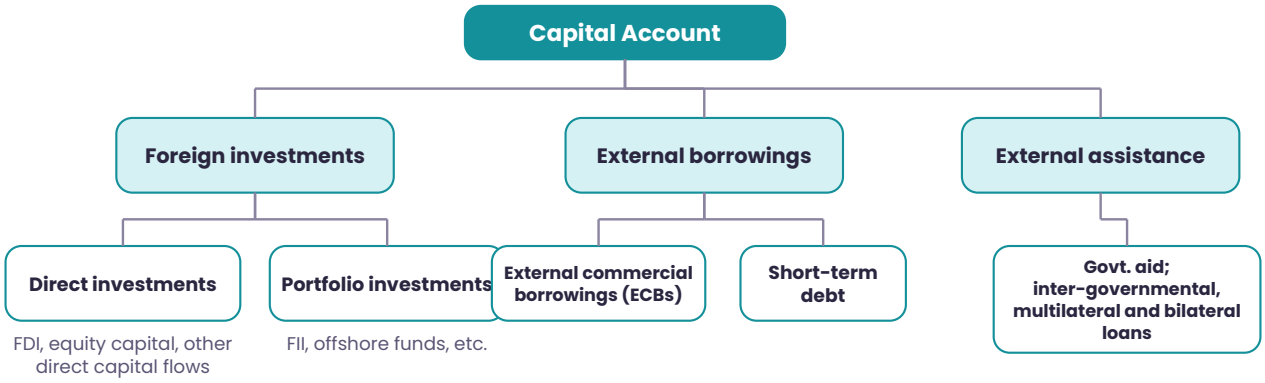
BALANCE OF PAYMENTS

Capital Account

16

Foreign investments, external borrowings and external assistance.

Capital account tree



Capital outflows or 'payments' = debit items

Capital inflows or 'receipts' = credit items

Change in forex reserves is also included

Capital account balance

Capital A/C surplus: Receipts are greater than expenditure (payments)

Capital A/C deficit: Receipts are less than expenditure (payments)

NOTE

FDI = direct investment (equity capital, direct capital flows). FII = portfolio investment (offshore funds). Both are foreign investment, but FDI means direct control or ownership.

BALANCE OF PAYMENTS | FOREIGN EXCHANGE

BoP Imbalances and Foreign Exchange

17

How the two accounts balance, and what moves the exchange rate.

BoP imbalances

BoP = Current A/C balance ① + Capital A/C balance ②

BoP has to be 0: if ① is greater than 0 ⇒ ② is less than 0; if ① is less than 0 ⇒ ② is greater than 0

If ① cannot offset ② ⇒ Δ Reserves

The RBI's foreign exchange reserves change to balance the BoP.

Nominal exchange rate (NER)

NER = domestic currency ÷ foreign currency (₹ / $)

NER ↑ ⇒ domestic currency ↓ (depreciation of ₹), foreign currency ↑ (appreciation of $)

NER ↓ ⇒ domestic currency ↑ (appreciation of ₹), foreign currency ↓ (depreciation of $)

Real exchange rate (RER)

$RER = NER \times (P_f \div P)$

P_f = price in foreign country | P = price in domestic country

QUICK MEMORY

More rupees needed for one dollar means the rupee has become weaker: NER goes up and the rupee depreciates.